

## SIMATS ENGINEERING ASSESSMENT TOOL 2

**COURSE NAME: COMPUTER NETWORKS**  
**COURSE CODE: CSA07**

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### COURSE OUTCOME COVERED

**CO4:**

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards  
(BL2,BL6)

Assessment Tool Weightage: 30%

82%  
\$

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Annexure A

### Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or

tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then

connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

|                | A | B | C | D | E | F | G | H   | I   |      |
|----------------|---|---|---|---|---|---|---|-----|-----|------|
| Q <sub>1</sub> | 3 | 3 | 4 | 2 | 2 | 2 | 1 | 0.5 | 1   | 18.5 |
| Q <sub>2</sub> | 2 | 4 | 3 | 2 | 2 | 1 | 1 | 1   | 0.5 | 16.5 |
| Q <sub>3</sub> | 3 | 4 | 2 | 1 | 1 | 1 | 1 | 1   | 0.5 | 14.5 |
| Q <sub>4</sub> | 2 | 3 | 4 | 1 | 2 | 2 | 1 | 1   | 1   | 17   |
| Q <sub>5</sub> | 3 | 2 | 4 | 1 | 2 | 1 | 1 | 0.5 | 1   | 15.5 |

82

## RUBRICS FOR CALCULATION

|   | Criteria                     | Weightage | Exemplary (4)                                                                | Proficient (3)                            | Developing (2)                                | Beginning (1)                                  |
|---|------------------------------|-----------|------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------|------------------------------------------------|
| A | Problem Analysis             | 15%<br>3  | Clearly identifies and deeply analyzes the industry problem with all factors | Good understanding with minor gaps        | Basic understanding; some key aspects missing | Poor or incorrect understanding of the problem |
| B | Solution Design              | 20%<br>4  | Innovative, practical, and industry-relevant solution                        | Effective solution with minor limitations | Partially suitable solution                   | Inappropriate or unclear solution              |
| C | Technical Implementation     | 20%<br>4  | Fully functional, accurate, and well-executed implementation                 | Mostly functional with minor errors       | Partially implemented solution                | Incomplete or non-functional implementation    |
| D | Use of Tools & Technologies  | 10%<br>2  | Appropriate and advanced use of relevant tools and technologies              | Correct use of tools                      | Limited or basic use of tools                 | Incorrect or no use of tools                   |
| E | Problem-Solving Approach     | 10%<br>2  | Logical, systematic, and well-structured approach                            | Mostly logical approach                   | Somewhat disorganized approach                | No clear approach                              |
| F | Innovation & Creativity      | 10%<br>2  | Highly creative and original solution                                        | Some creativity                           | Limited originality                           | No creativity                                  |
| G | Testing & Validation         | 5%<br>1   | Thorough testing with real-world scenarios                                   | Adequate testing                          | Minimal testing                               | No testing                                     |
| H | Documentation & Presentation | 5%<br>1   | Clear, professional, and well-documented                                     | Good documentation                        | Basic documentation                           | Poor or no documentation                       |
| I | Teamwork / Collaboration     | 5%<br>1   | Excellent coordination and contribution (if team-based)                      | Good collaboration                        | Limited participation                         | No collaboration or contribution               |

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(BL2,BL6)

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### Annexure A

### Industry Problem-Solving Task

1.A-star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

## **Answers:**

### **Star Topology for a 20-Person Startup**

A 20-person startup needs a reliable computer network to connect employees' computers, printers, VoIP phones, laptops, and other network devices. Since the company may grow in the future, the network should be easy to maintain, troubleshoot, and expand. To solve this requirement, a star topology is a suitable networking solution because all devices are connected to a central switch.

In this network, a 24-port managed switch is placed at the center. The switch provides enough ports to connect the 20 employees' computers along with printers, VoIP phones, and other devices. Extra ports are available for connecting additional devices when the company expands. A managed switch also allows the network administrator to monitor and control network traffic.

Each computer and other wired device is connected to the central switch using Cat6 UTP cables. Cat6 cables provide high-speed network communication and are suitable for office environments. The cables are arranged from each individual device to the central switch, forming a star-shaped network structure.

The central switch is connected to a router/firewall. The router provides Internet connectivity to all devices in the office. It can also provide DHCP, which automatically assigns IP addresses to computers, and NAT, which allows multiple internal devices to share a public Internet connection. The firewall helps protect the internal network from unwanted external traffic.

A Wi-Fi access point can also be connected to the switch. It provides wireless connectivity for laptops, smartphones, tablets, and other mobile devices. This allows employees to access company network resources and the Internet without requiring a physical cable connection.

The star topology provides an important fault-isolation mechanism. For example, if the cable connecting Employee 5's computer to the switch becomes damaged, only Employee 5 loses network connectivity. The other 19 employees can continue their work normally. This makes it easier for the network administrator to identify and repair individual connection problems.

Another important advantage is easy maintenance. Since all connections meet at the central switch, the administrator can easily identify which port belongs to a particular device. Network monitoring

and troubleshooting can therefore be performed more efficiently. Devices can also be replaced without affecting the entire network.

The topology is also scalable. If the startup hires more employees, additional computers can be connected to the available switch ports. If all ports become occupied, another switch can be added to increase the network capacity. Therefore, the network can grow along with the organization.

Security can also be improved using the managed switch and router/firewall. The administrator can configure appropriate access controls, separate different types of devices, monitor network traffic, and restrict unauthorized access. For example, employee computers, guest Wi-Fi devices, and office equipment can be managed according to their requirements.

From a cost perspective, star topology may require more cables than bus topology because every device needs an individual connection to the switch. However, the additional cabling is justified because the network provides better reliability, easier troubleshooting, and simpler future expansion.

For example, consider a situation where one employee suddenly loses Internet access. In a star topology, the administrator can check that employee's cable, switch port, and network configuration without shutting down the entire network. If the problem is caused by a faulty cable, replacing that cable can restore the connection while the rest of the office continues operating.

Thus, the proposed star topology solves the startup's networking requirements by providing centralized management, reliable communication, easy fault detection, improved maintenance, and future scalability. It is an effective and practical network design for a small and growing organization.

**2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.**

**Answers:**

### **Mesh Topology for a Bank**

A bank requires a highly reliable and secure network because its banking services must be available almost continuously. Network downtime can interrupt ATM services, online banking, branch communication, financial transactions, and internal banking applications. Therefore, a network topology with multiple backup paths is required. A full or partial mesh topology is an appropriate solution for this problem.

In a full mesh topology, every important network device is directly connected to every other device. This creates multiple communication paths between routers, core switches, firewalls, and servers. If one connection fails, network traffic can use another available path. However, because a full mesh requires a large number of connections, it can be expensive and difficult to manage in a large banking environment.

Therefore, banks commonly use a partial mesh topology for the core network. In this design, important core switches, routers, and firewalls have multiple connections between them. The distribution and access layers can also have redundant uplinks to two or more core switches. This ensures that the failure of one switch or link does not disconnect the entire network.

For Internet connectivity, the bank can use two enterprise-grade routers connected to different Internet service providers. Routing protocols such as BGP can be used to provide redundant Internet connections. If one Internet connection becomes unavailable, traffic can be redirected through the second connection.

The connections between important network devices can use fiber-optic cables. Single-mode fiber can be used for long-distance connections between different buildings, while multi-mode fiber such as OM4 can be used for shorter connections within a data center. Fiber provides high bandwidth, low latency, and resistance to electromagnetic interference, which is useful for high-speed banking transactions.

Network redundancy should not be limited to cables and switches. The bank can also use redundant power supplies, UPS systems, backup servers, and RAID storage. These additional components reduce the risk of service interruption caused by power failure or hardware failure.

Routing protocols such as OSPF or BGP can help identify available network paths and redirect traffic when a connection fails. For example, if one fiber link between two core switches fails, traffic can use another available path instead of stopping completely.

Security is also important in a banking network. Firewalls, access controls, network segmentation, monitoring systems, and secure communication protocols can be implemented to protect sensitive customer and financial information.

The main advantage of mesh topology is high reliability and fault tolerance. A single link or device failure does not necessarily bring down the complete network because alternative communication paths are available. This helps the bank maintain continuous services and reduce network downtime.

Therefore, a partial mesh topology with redundant core devices, multiple Internet connections, fiber-optic links, backup power, and reliable routing protocols provides a strong solution for a bank. It supports high availability, fault tolerance, fast recovery, and secure communication, making it suitable for critical banking operations.

**3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.**

**Answers:**

### **Bus Topology for a Small Classroom Lab**

A small classroom computer lab may need a network to connect several computers for activities such as Internet browsing, file sharing, and accessing learning resources. Since the classroom has a limited budget, the network should be simple and inexpensive to install. A bus topology can be considered as a low-cost solution because multiple computers share a common communication path.

In a traditional bus topology, all computers are connected to a single backbone cable. This reduces the amount of cabling and networking equipment required. A classic implementation can use a 50-ohm RG-58 coaxial cable as the main communication line. Each computer is connected to the backbone using a T-connector and BNC connector.

In the traditional coaxial arrangement, 50-ohm terminators are placed at both ends of the backbone cable. These terminators prevent signal reflections and help maintain proper communication along the cable. This design does not require a dedicated central switch, which reduces the initial hardware cost.

For a modern classroom network, however, Ethernet networks normally use switches rather than a true coaxial bus. A low-cost 8-port or 16-port unmanaged switch with Cat5e or Cat6 cables can provide a similar simple and economical classroom setup. Each workstation can be connected to the switch using inexpensive UTP patch cables.

For example, if a classroom has 10 computers, the network can connect all the computers using affordable networking equipment. The teacher's computer can also provide shared files or other resources using the built-in file-sharing features of Windows or Linux. Therefore, a separate dedicated server may not be necessary for basic classroom activities.

The main advantage of this solution is low cost and simple installation. It requires less specialized equipment and can be suitable for a small environment where network traffic is relatively low. Students can use the network for Internet browsing, sharing small files, and accessing educational resources.

However, the traditional bus topology has some important limitations. Since multiple computers share the same communication medium, heavy network traffic can cause collisions and reduced performance. More importantly, if the main backbone cable is damaged, communication for multiple connected computers can be disrupted.

For a supervised classroom with light network usage, these limitations may be acceptable when cost is the main consideration. The network administrator or teacher can also monitor the connections and quickly identify basic problems.

Therefore, a low-cost bus-based or simple switched Ethernet setup can be considered for a small classroom lab. It provides an economical solution for basic networking requirements while avoiding unnecessary investment in expensive networking equipment.

**4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide**

**Answers:**

### **Hybrid Topology for a Large University Campus**

A large university campus may have thousands of students, faculty members, and administrative staff using the network every day. The campus can include lecture halls, computer laboratories, libraries, hostels, administrative offices, research centers, and other facilities. Since these buildings have different networking requirements, using only one network topology would not be efficient. Therefore, a hybrid topology is a suitable solution because it combines multiple topology designs according to the requirements of different areas.

At the campus backbone level, a high-speed fiber-optic ring or partial mesh topology can be implemented. Core routers and multilayer switches are connected using high-speed fiber links. Single-mode fiber can be used for long-distance connections between buildings, while suitable high-speed fiber infrastructure can provide 10 Gbps or higher connectivity where required. The redundant arrangement provides alternative paths for network traffic when a link fails.

Each university building can have a distribution switch that connects the building to the campus core. The distribution switch can have dual fiber uplinks to different core devices. This provides redundancy because if one fiber link fails, the second connection can continue carrying network traffic.

Inside each building, a star topology can be used. The distribution switch connects to floor-level access switches, and these access switches connect individual computers, printers, IP phones, and other devices. Cat6 Ethernet cables can be used for wired connections. This arrangement makes the network easier to manage because devices on each floor can be controlled through the corresponding access switch.

For individual computer laboratories, smaller star, tree, or bus-like arrangements can be selected depending on the number of computers, available equipment, and budget. For example, a computer lab can use an access switch connected to multiple student computers. This provides an organized and manageable local network.

Wireless connectivity is also essential on a university campus. Wireless access points can be installed throughout lecture halls, libraries, hostels, and common areas. Access points can be connected to network switches using Power over Ethernet (PoE), allowing both network connectivity and electrical power to be supplied through the Ethernet cable. This reduces the need for separate power connections at every access point.

The hybrid design follows a hierarchical three-layer architecture consisting of the core, distribution, and access layers. The core layer provides high-speed communication between buildings. The distribution layer manages traffic and provides redundancy for individual buildings.

One major advantage of this design is scalability. If the university constructs a new building, a new distribution switch can be installed and connected to the existing campus core using fiber-optic links. The existing buildings do not need to be completely redesigned. Similarly, additional computers and access points can be added to the access layer when required.

If one backbone connection fails, an alternative path can be used to maintain communication. If an access switch fails, the problem can generally be isolated to the devices connected to that switch rather than affecting the entire campus.

For example, if the fiber connection between the library and one core switch fails, the library's secondary fiber connection can continue providing network access. At the same time, students in other buildings can continue using the campus network normally. Therefore, a hybrid topology provides an effective solution for a large university campus. It combines redundancy at the backbone level, centralized management at the building level, simple device connectivity at the access level, wireless connectivity, and campus-wide scalability.

**5.Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.**

**Answers:**

### **Designing a Network for a Company with Multiple Departments**

A company with multiple departments such as HR, Finance, Sales, and IT requires a well-organized network to allow employees to communicate and share resources while protecting sensitive departmental information. A single flat network can create security and performance problems because all devices belong to the same broadcast domain. Therefore, a hierarchical network architecture using switches, routers, VLANs, and proper IP addressing can be implemented.

At the access layer, each department can have a dedicated Layer 2 switch. Computers, printers, IP phones, and other devices are connected to these switches. For example, the HR department can use one switch, finance another switch, and Sales and IT can have their own switches. This provides organized connectivity and makes it easier for the network administrator to manage individual departments.

To logically separate the departments, VLANs (Virtual Local Area Networks) are configured. For example, VLAN 10 can be assigned to HR, VLAN 20 to Finance, VLAN 30 to Sales, and VLAN 40 to IT. Devices belonging to the same VLAN can communicate within their department, while communication between different VLANs requires routing. This separation improves security and reduces unnecessary broadcast traffic.

A Layer 3 switch can be placed at the core or distribution layer to perform inter-VLAN routing. It allows authorized communication between different departments. For example, the Finance department may need to access a database server located in the IT department, while ordinary users may not be allowed to access sensitive Finance systems. Access policies can therefore be applied to control communication between VLANs.

The Layer 3 switch can be connected to an edge router or firewall, which provides connectivity to external networks and the Internet. The firewall can control incoming and outgoing traffic and help protect the company's internal resources from unauthorized access. Network security rules can be configured according to the requirements of each department.

Proper IP addressing and subnetting should also be implemented. Each VLAN can be assigned a separate IP subnet. For example, HR may use 192.168.10.0/24, Finance may use 192.168.20.0/24, Sales may use 192.168.30.0/24, and IT may use 192.168.40.0/24. This makes network management easier and allows administrators to identify devices based on their department.

Redundant links and backup paths can be included to improve reliability. Important switches can have multiple connections, so if one link fails, another link can continue carrying network traffic. This reduces downtime and ensures that employees can continue working even when a network component experiences a failure.

For example, if the Finance department needs to access an application server located in the IT department, the Finance computer sends the traffic through its VLAN gateway. The Layer 3 switch identifies the destination network and routes the packet to the appropriate VLAN. Security rules can determine whether the communication is permitted.

The proposed hierarchical design provides several advantages. It improves network performance, security, fault isolation, and manageability. VLANs prevent unnecessary broadcast traffic from spreading across the entire company, while inter-VLAN routing allows controlled communication between departments. The architecture is also scalable because additional departments, users, switches, and VLANs can be added as the company grows.

Therefore, a hierarchical network using Layer 2 access switches, VLANs, a Layer 3 switch, routers/firewalls, redundant links, and proper IP addressing provides an efficient and secure solution for a multi-department company. It supports current business requirements while allowing the network to expand easily in the future.

## RUBRICS FOR CALCULATION

| Criteria                     | Weightage | Exemplary (4)                                                                | Proficient (3)                            | Developing (2)                                | Beginning (1)                                  |
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